Intellectual Developmental Disorder with Speech Delay, Autism, and Dysmorphic Facies (IDDSADF) — Comprehensive Disease Characteristics Report

Disease: Intellectual Developmental Disorder with Speech Delay, Autism, and Dysmorphic Facies (IDDSADF) **MONDO ID:** MONDO:0032864 · **OMIM phenotype:** #618672 · **Causal gene:** *CNOT3* (19q13.42) **Category:** Mendelian, autosomal dominant

Summary

Intellectual Developmental Disorder with Speech Delay, Autism, and Dysmorphic Facies (IDDSADF) is a rare autosomal-dominant Mendelian neurodevelopmental disorder caused by heterozygous, predominantly *de novo* variants in **CNOT3** (CCR4-NOT transcription complex subunit 3; HGNC:7879; NCBIGene:4849; ENSG00000088038; UniProt O75175; 19q13.42). The disorder was first delineated as a discrete entity by Martin et al. (2019) in 16 probands ascertained through the Deciphering Developmental Disorders (DDD) study, each carrying *de novo CNOT3* variants PMID: 31201375. The phenotype combines global developmental delay/intellectual disability with speech delay, hypotonia, motor delay, autistic/behavioral features, relatively small stature, and a mild, non-recognizable facial dysmorphism; seizures are uncommon.

CNOT3 is a scaffold subunit of the CCR4-NOT complex, the principal cytoplasmic mRNA deadenylase that shortens poly(A) tails to trigger mRNA decay and that also contributes to transcription and protein ubiquitination. The gene is among the most mutation-intolerant in the human genome (gnomAD pLI = 1.0; LOEUF = 0.07; missense z = 4.65), and the balance of truncating and missense alleles, together with the authors' own conclusion, indicates that **haploinsufficiency** is the most likely mechanism. Reduced CNOT3 dosage is inferred to impair poly(A)-tail shortening and mRNA decay, dysregulating the post-transcriptional gene-expression programs that govern cell differentiation and identity — a role demonstrated

across embryonic stem cells, pancreatic β-cells, hepatocytes, cardiomyocytes, and innate lymphoid cells in model systems. How this general post-transcriptional defect is translated into the specific neurodevelopmental phenotype in humans has not yet been directly demonstrated and remains the central knowledge gap.

Clinically, there is no targeted therapy. Diagnosis is molecular, achieved primarily through trio exome or genome sequencing; there is no biochemical biomarker and brain MRI findings are non-specific. Management is entirely supportive — early intervention, speech/occupational/physical therapy, special education, and behavioral support. An important pleiotropic observation is that *de novo CNOT3* variants have also been associated with moyamoya angiopathy, extending the clinical spectrum into later-onset cerebrovascular disease and warranting awareness in long-term follow-up.

> **Note on provenance / a corrected mis-identification.** Early in this investigation the MONDO ID was provisionally matched to KDM6B-related disorder (OMIM #618906). This was **retracted** upon ontology cross-referencing: MONDO:0032864 maps unambiguously to OMIM:618672 / MedGen C5231456 / GARD:0027947, and the causal gene is **CNOT3**, not KDM6B. All conclusions in this report refer to the CNOT3 disorder.

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1. Disease Information

Overview. IDDSADF is a monogenic neurodevelopmental disorder characterized by global developmental delay and intellectual disability, prominent speech delay, autistic/behavioral features, hypotonia, relatively small stature, and mild facial dysmorphism without a readily recognizable gestalt. It is a "developmental disorder" in the DDD-study sense — a congenital-onset, non-progressive condition of brain development.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0032864
OMIM (phenotype)	#618672
OMIM (gene *CNOT3*)	604910
MedGen	C5231456 (also 1684848)
GARD	0027947
Gene (NCBI)	4849
HGNC	7879
Ensembl	ENSG00000088038
UniProt	O75175
Cytoband	19q13.42

Synonyms / alternative names. IDDSADF; CNOT3-related neurodevelopmental disorder; CNOT3-associated developmental disorder; intellectual developmental disorder with speech delay, autism, and dysmorphic facies.

Source of information. The entry is derived from **aggregated disease-level resources** (OMIM, Orphanet, HPO, MONDO) that themselves summarize **individual-patient data** from published cohorts — primarily the 16-proband DDD cohort of Martin et al. 2019 PMID: 31201375 plus subsequent case reports (Meyer et al. 2020 PMID: 32720325).

2. Etiology

Disease causal factor — genetic. IDDSADF is caused by heterozygous, predominantly *de novo* pathogenic variants in *CNOT3*. There is no known environmental, infectious, or mechanistic (non-genetic) cause.

> *Martin et al. 2019:* "de novo variants in CNOT3 have emerged as a newly recognised cause of a developmental disorder." PMID: 31201375

Genetic risk factors. The causal variants are the risk factor. *CNOT3* is extremely intolerant to loss-of-function and missense variation (gnomAD pLI = 1.0; LoF o/e = 0.022, LOEUF = 0.07; missense z = 4.65), so a single damaging allele is sufficient to cause disease (see §4, §6). No independent susceptibility loci or modifier genes have been established.

Environmental risk factors. None identified. Because most variants arise *de novo*, advanced parental age is a plausible but unproven generic contributor to *de novo* mutation rate (not specifically demonstrated for *CNOT3*).

Protective factors. None identified — no protective alleles or environmental protective exposures are known for this rare Mendelian disorder.

Gene–environment interactions. None documented. The disorder is essentially fully genetically determined given a pathogenic allele; variable expressivity (see §9) is more likely attributable to genetic background/stochastic developmental factors than to defined environmental modifiers.

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3. Phenotypes

Curated HPO annotations for OMIM:618672, with cohort frequencies drawn from Martin et al. 2019 (n = 16) PMID: 31201375:

Phenotype	HPO term	Type	Frequency (cohort)
Intellectual disability	HP:0001249	Cognitive/ behavioral	Core / consistent
Global developmental delay	HP:0001263	Developmental	Core / consistent
Delayed ability to walk	HP:0031936	Motor	12/15
Hypotonia	HP:0001252	Neuromuscular sign	10/16
Autistic behavior	HP:0000729	Behavioral	7/16
Prominent forehead	HP:0011220	Dysmorphism	5/16
Malar flattening	HP:0000272	Dysmorphism	4/16
Short philtrum	HP:0000322	Dysmorphism	4/16
Strabismus	HP:0000486	Ophthalmologic	4/16
Deeply set eye	HP:0000490	Dysmorphism	4/16
Tapered finger	HP:0001182	Skeletal/ dysmorphism	3/16
Hypoplasia of corpus callosum	HP:0002079	Neuroimaging	2/12
Speech delay	HP:0000750	Developmental	Core (named in disease title)
Short stature / small stature	HP:0004322	Growth	Frequent ("relatively small stature")

Phenotype characteristics. - **Age of onset:** infantile/congenital (HP:0003593) — developmental delay is evident in the first years of life. - **Severity:** variable; intellectual disability ranges from mild to moderate–severe across the cohort. - **Progression:** stable/non-progressive (a static encephalopathy of developmental origin), not neurodegenerative. - **Facial dysmorphism:** mild and inconsistent — notably **no easily recognizable facial gestalt**; recurring features include anteverted nares, thin upper lip, and low-set eyebrows.

> *Martin et al. 2019:* "The most consistent phenotypic features seen in subjects with de novo variants in CNOT3 were hypotonia, relatively small stature, developmental delay, behavioural problems and intellectual disability. There is no easily recognisable facial phenotype, but some common dysmorphic features such as anteverted nares, thin upper lip and low set eyebrows were shared among some of the probands." PMID: 31201375

Quality-of-life impact. Intellectual disability, speech delay, and autistic/behavioral features are the dominant drivers of long-term functional limitation, affecting communication, education, independent living, and social participation. Hypotonia and motor delay add early functional burden. No disease-specific QoL instruments (EQ-5D, SF-36, PROMIS) have been applied to this rare disorder.

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4. Genetic / Molecular Information

Causal gene. *CNOT3* — CCR4-NOT transcription complex subunit 3 (aliases NOT3, NOT3H, LENG2). Gene OMIM 604910; HGNC:7879; NCBIGene:4849; ENSG00000088038; UniProt O75175; cytoband **19q13.42** (GRCh38 chr19:54,137,749–54,155,681).

Pathogenic variants. - **Variant classes:** both protein-truncating variants (nonsense, frameshift, splice-site) and missense. In the defining cohort, **8/16 probands had protein-truncating variants and 8/16 had missense variants**; three of the missense cases shared an amino-acid substitution at the **same residue**, suggesting a functionally critical position. - **Classification (ACMG/AMP):** ClinVar (queried 2026-10) lists **530 CNOT3 variant records — 183 pathogenic, 44 likely pathogenic, and 422 of uncertain significance (VUS)** (counts overlap across submissions/categories). Disease-causing alleles are heterozygous. - **Allele frequency:** pathogenic alleles are absent/ultra-rare in gnomAD, consistent with strong negative selection and a largely *de novo* origin. - **Somatic vs germline: germline** (constitutional), overwhelmingly *de novo*; rare parent-to-child transmission is documented (see §9). - **Functional consequence: loss of function / haploinsufficiency.** Missense alleles at critical residues are inferred to be damaging/hypomorphic rather than gain-of-function.

> *Martin et al. 2019:* "Haploinsufficiency appears to be the most likely mechanism of action, with eight cases found to have protein-truncating variants. Of the other eight cases (all missense variants), three share an amino acid substitution at the same position which may therefore represent an important functional domain." PMID: 31201375

Constraint metrics (gnomAD).

Metric	*CNOT3* value	Interpretation
pLI	1.0	Fully LoF-intolerant
LoF o/e (LOEUF)	0.022 (0.07)	2 observed vs 89.4 expected LoF
LoF z	7.84	Extreme LoF constraint
Missense o/e	0.657	Depleted for missense
Missense z	4.65	Strong missense constraint

These place *CNOT3* among the most constrained genes in the genome and directly support a dosage-sensitive, haploinsufficiency mechanism.

Modifier genes. None established. Variable expressivity within families (§9) implies genetic/stochastic modifiers whose identity is unknown.

Epigenetic information. No disease-specific DNA-methylation episignature has been reported for CNOT3-IDDSADF (unlike several chromatin-gene NDDs). Not applicable/unknown.

Chromosomal abnormalities. Most cases are SNV/indel; chromosomal microarray can detect 19q13.42 copy-number changes, but large structural rearrangements are not the typical mechanism.

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5. Environmental Information

Not applicable. IDDSADF is a monogenic disorder with no established environmental, lifestyle, toxic, or infectious contributors. No gene–environment interaction has been demonstrated.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A heterozygous *de novo* (rarely inherited) **loss-of-function or damaging missense variant arises in *CNOT3*** (19q13.42). *[demonstrated — variant data, Martin 2019]*
2. This **reduces functional CNOT3 protein to ~50%** (haploinsufficiency), because *CNOT3* is dosage-sensitive and LoF-intolerant (pLI = 1, LOEUF = 0.07). *[inferred from constraint + truncating/missense split]*
3. Reduced CNOT3 **impairs assembly/scaffolding of the CCR4-NOT deadenylase complex**, of which CNOT3 is a NOT-module scaffold subunit. *[demonstrated — CNOT3 is a defined scaffold subunit]*
4. Impaired CCR4-NOT function **reduces mRNA poly(A)-tail shortening (deadenylation)**, the rate-limiting step of bulk cytoplasmic mRNA decay. *[demonstrated — complex biochemistry]*
5. Defective deadenylation **stabilizes and dysregulates target mRNAs**, perturbing the post-transcriptional gene-expression programs that control **cell differentiation, maturation, and identity**. *[demonstrated in multiple cell types — ESC, β-cell, hepatocyte, ILC2]*
6. During neurodevelopment, this dysregulation **disrupts the timing and fidelity of neuronal differentiation and brain patterning**. *[inferred — not yet directly shown in human/mouse neurons]*
7. The result is **static, congenital brain dysfunction** manifesting as intellectual disability, speech delay, hypotonia, autistic/behavioral features, and mild dysmorphism. *[demonstrated — clinical cohort]*

Branch (pleiotropy): independently, *de novo CNOT3* variants **predispose to moyamoya angiopathy**, a later-onset large-artery occlusive cerebrovascular disease, implicating CNOT3-dependent post-transcriptional control in vascular as well as neural development. *[demonstrated — Pinard 2020]*

Detail by category

Molecular pathway. The core lesion is in the **CCR4-NOT mRNA-deadenylation pathway** (cytoplasmic mRNA turnover / post-transcriptional regulation), not a classical signaling cascade (Wnt/MAPK/mTOR). CNOT3 also contributes to transcription and protein ubiquitination.

> *Kuba 2018:* "deadenylation is mostly executed by the CCR4-NOT complex. CCR4-NOT complex is a multi-subunit protein complex, which controls gene expression in the levels from transcription through mRNA deadenylation and protein ubiquitination. We had previously identified CNOT3, a scaffold subunit of the CCR4-NOT complex." PMID: 29526923

Cellular processes. Loss of cell-identity maintenance and failure of proper maturation/differentiation; in some contexts, de-repression of senescence programs. CNOT3 knockdown stabilizes *CDKN1A*(p21)/*TP53* transcripts and induces cellular senescence hallmarks PMID: 39823890.

Protein dysfunction. Truncating variants delete functional domains / trigger nonsense-mediated decay (loss of function); recurrent same-residue missense variants likely disrupt a critical functional domain (hypomorphic loss of function). Net effect: reduced deadenylase scaffold function.

Metabolic / biochemical abnormalities. No specific enzyme deficiency or metabolite is measurable clinically; the "biochemical" defect is global, at the level of mRNA poly(A) metabolism rather than intermediary metabolism — so there is no diagnostic biomarker.

Immune involvement. Not a feature of the human disorder, though model work shows CNOT3 shapes innate lymphoid cell (ILC2) differentiation by destabilizing *Tbx21*/*Rorc* PMID: 42118148 — illustrating the breadth of CNOT3's identity-controlling function rather than a disease immunophenotype.

Molecular profiling. Functional studies (model organisms, see §15) show that *Cnot3* deletion increases mRNA stability of differentiation regulators and derails lineage programs; no human patient-tissue transcriptomic/proteomic signature has been published.

Evidence that CNOT3-dependent deadenylation controls cell identity:

> *Mostafa et al. 2020:* "CNOT3, a CCR4-NOT deadenylase complex subunit, is dysregulated in islets in diabetic db/db mice, and that it is essential for murine β cell maturation and identity." PMID: 32859966

GO / CL / UBERON suggestions. - GO biological process: mRNA deadenylation (GO:0000289); nuclear-transcribed mRNA poly(A) tail shortening (GO:0000290); regulation of mRNA stability (GO:0043488); nervous system development (GO:0007399); cell differentiation (GO:0030154). -

GO cellular component: CCR4-NOT complex (GO:0030014); cytoplasm (GO:0005737); nucleus (GO:0005634). - CL cell types (inferred): neuron (CL:0000540); glutamatergic/GABAergic neurons; neural progenitor/stem cell (CL:0000047). - UBERON: brain (UBERON:0000955); cerebral cortex (UBERON:0000956); corpus callosum (UBERON:0002336).

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7. Anatomical Structures Affected

Organ level. Primary organ is the **brain** (central nervous system; body system = nervous system). Secondary/associated involvement: ophthalmologic (strabismus), musculoskeletal (hypotonia, tapered fingers, small stature), craniofacial skeleton (mild dysmorphism). The pleiotropic branch involves the **cerebral arteries** (moyamoya angiopathy).

Tissue / cell level. Affected tissue is **neural (nervous) tissue**. Inferred target populations: neurons and neural progenitor cells during corticogenesis; in the moyamoya branch, arterial **vascular smooth muscle / endothelium**. No specific neuronal subtype has been proven as the locus of disease.

Subcellular level. The molecular machinery localizes to the **cytoplasm** (site of CCR4-NOT–mediated deadenylation; GO:0005737) with additional **nuclear** functions (GO:0005634). The relevant subcellular process is cytoplasmic mRNA turnover rather than organelle pathology.

Localization & lateralization. Brain involvement is **bilateral/global**. Structural neuroimaging is non-specific — occasional thin/hypoplastic corpus callosum (UBERON:0002336), enlarged cisterna magna, and delayed myelination. UBERON terms: brain (UBERON:0000955), cerebral cortex (UBERON:0000956), corpus callosum (UBERON:0002336), cerebral artery (UBERON:0001629).

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8. Temporal Development

Onset. Congenital/infantile; developmental delay and hypotonia are typically apparent in the first year of life. Onset pattern is **insidious/chronic** (developmental), not acute.

Progression. Static (non-progressive) encephalopathy — intellectual disability and core features are stable over time rather than degenerative. Disease duration is **chronic/lifelong**.

Patterns. No remission or relapsing course for the neurodevelopmental features. The **critical period** for intervention is early childhood (early-intervention and therapy windows for language, motor, and behavioral development). The **moyamoya branch** introduces a distinct later-onset, potentially progressive cerebrovascular time-course requiring separate surveillance.

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9. Inheritance and Population

Epidemiology. Ultra-rare. Precise prevalence/incidence are **not established**; the literature comprises on the order of dozens of reported individuals (16 in the defining DDD cohort plus subsequent cases). No registry-based prevalence figures are available.

Inheritance pattern. Autosomal dominant (HP:0000006). The overwhelming majority of cases are **de novo**; Meyer et al. 2020 reported the **first two unrelated families with parent-to-child transmission**, establishing that the allele can be inherited and showing **substantial intra-familial phenotypic variability (variable expressivity)**.

> *Meyer et al. 2020:* "We report for the first time autosomal dominant transmissions of the CNOT3-associated developmental disorder in two unrelated families. The clinical characteristics in our patients match the IDDSADF features reported so far and suggest substantial variability of the phenotype within the same family." PMID: 32720325

Penetrance / expressivity. Penetrance appears high for a neurodevelopmental phenotype but **expressivity is variable**, even within families. Genetic anticipation: not applicable (not a repeat-expansion disorder). Germline mosaicism: plausible for *de novo* disorders but not specifically documented for *CNOT3*. Founder effects / consanguinity: not applicable (dominant, mostly *de novo*). Carrier frequency: not applicable (no asymptomatic-carrier recessive state; the disease is dominant).

Population demographics. No ethnic predilection, geographic clustering, or variant-specific geographic distribution is known. Sex ratio is not established (no strong sex bias reported). Age distribution: diagnosed in childhood.

Pleiotropy. *De novo CNOT3* variants also predispose to **moyamoya angiopathy**, grouping *CNOT3* with chromatin/developmental NDD genes (*CHD4*, *SETD5*) whose pleiotropy extends into cerebrovascular disease.

> *Pinard et al. 2020:* "We identified four de novo variants in three genes not previously associated with MMA: CHD4, CNOT3, and SETD5… these data expand the spectrum of phenotypic pleiotropy due to alterations of CHD4, CNOT3, and SETD5 beyond DD to later onset disease in the cerebrovascular arteries." PMID: 31474762

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10. Diagnostics

Diagnostic approach — molecular. Diagnosis rests on identifying a heterozygous pathogenic/likely-pathogenic *CNOT3* variant. The disorder was itself discovered through exome sequencing in the DDD study.

> *Martin et al. 2019:* "As a result of exome-based sequencing work performed by the DDD study, de novo variants in CNOT3 have emerged as a newly recognised cause of a developmental disorder." PMID: 31201375

Modality	Utility for IDDSADF
Trio exome sequencing (WES)	**Primary/highest-yield** — detects *de novo* SNV/indel; trio design confirms *de novo* status
Genome sequencing (WGS)	Alternative; adds non-coding/structural detection
ID/ASD/epilepsy multigene panels (incl. *CNOT3*)	Targeted alternative
Single-gene *CNOT3* testing	For confirmation or familial cascade testing
Chromosomal microarray (CMA)	Detects 19q13.42 CNV; most cases are SNV/indel (lower yield)
Karyotype / FISH	Low yield (not a translocation disorder)
mtDNA / repeat-expansion testing	Not applicable

Clinical / laboratory tests. No specific blood, urine, or enzyme biomarker exists. Routine metabolic workup is non-diagnostic.

Imaging. Brain MRI is **non-specific**: occasional thin/hypoplastic corpus callosum, enlarged cisterna magna, delayed myelination. In patients with cerebrovascular symptoms, MRA/angiography should be considered to evaluate for moyamoya.

Omics-based diagnostics. No validated RNA/proteomic/metabolomic/methylation diagnostic signature is currently available for CNOT3-IDDSADF.

Clinical criteria / differential diagnosis. There are no formal consensus diagnostic criteria; diagnosis is gene-based. The non-specific phenotype overlaps broadly with other monogenic intellectual-disability/autism syndromes (e.g., chromatin-regulator and transcription/mRNA-regulation NDDs), so molecular testing is required to distinguish them.

Screening. No population newborn or carrier screening exists or is indicated (dominant, mostly *de novo*, ultra-rare).

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11. Outcome / Prognosis

Survival / mortality. The core neurodevelopmental disorder is **not life-limiting** per se; no disease-specific mortality or reduced-life-expectancy data are established. Life expectancy is presumed near-normal in the absence of serious complications.

Morbidity / function. The principal long-term burden is **cognitive and communicative disability** — intellectual disability and speech delay affecting education, independence, and social participation — plus behavioral/autistic features. Hypotonia and motor delay contribute early functional impairment. No standardized QoL measures have been applied.

Disease course / complications. Core features are static. The most clinically significant potential complication is the pleiotropic **moyamoya cerebrovascular disease** (risk of TIA/stroke), which, where present, carries its own morbidity and warrants neurovascular surveillance and management.

Recovery potential. The developmental disability is permanent; however, **functional gains are achievable with early intervention and therapies**. Prognosis is otherwise variable, mirroring the variable expressivity of the phenotype.

Prognostic factors / biomarkers. No validated molecular prognostic biomarker. Genotype (truncating vs specific missense) has not been shown to reliably predict severity; variable expressivity is observed even within a family at a shared variant.

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12. Treatment

There is no targeted or disease-modifying therapy. Management is entirely **supportive and multidisciplinary**, directed at the symptoms.

Intervention	Purpose	NCIT suggestion
Early-intervention developmental programs	Maximize developmental trajectory	NCIT:C15317 (Rehabilitation)
Speech-language therapy	Address speech delay / communication	NCIT:C15381 (Speech Therapy)
Occupational therapy	Fine motor, daily-living skills	NCIT:C15349 (Occupational Therapy)
Physical therapy	Hypotonia, motor delay	NCIT:C15344 (Physical Therapy)
Special education / IEP support	Learning and cognition	—
Behavioral / ASD-directed support	Autistic/behavioral features	NCIT:C53555 (Behavioral Therapy)
Ophthalmology (strabismus)	Correct strabismus	—
Neurovascular management (if moyamoya)	Stroke prevention/ treatment	NCIT:C15329 (Surgical procedure, e.g., revascularization)

Pharmacotherapy. No gene-specific drug. Symptomatic psychotropic/behavioral medications may be used per standard ASD/ADHD/behavioral practice; no pharmacogenomic guidance is specific to *CNOT3*.

Advanced therapeutics (gene/cell/RNA/targeted/immuno). None approved or in trials for this disorder.

Experimental treatments. No disease-specific clinical trials (no NCT identifiers) are known for CNOT3-IDDSADF.

Treatment outcomes / strategy. No response-rate data; care follows general neurodevelopmental-disability pathways with individualized, combination (multidisciplinary) support. Personalized medicine is currently limited to accurate molecular diagnosis, genetic counseling, and surveillance for pleiotropic complications.

13. Prevention

- **Primary prevention:** Not possible for *de novo* disease. Where a variant is inherited (rare), **genetic counseling** informs recurrence risk (up to 50% transmission for an affected/carrier parent).
- **Secondary prevention:** Early developmental surveillance and **early-intervention therapy** to optimize outcomes once diagnosed.
- **Tertiary prevention:** Monitoring for and managing complications — notably **neurovascular surveillance for moyamoya** in symptomatic individuals, plus management of strabismus and behavioral issues.
- **Genetic screening / reproductive options:** For families with an identified variant, options include prenatal diagnosis and preimplantation genetic testing (PGT); cascade testing of at-risk relatives where transmission has occurred.
- **Immunization / public-health / environmental measures:** Not applicable.
- **Counseling:** Genetic counseling is central — explaining *de novo* origin, low but non-zero recurrence (germline mosaicism cannot be excluded), and variable expressivity.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CNOT3* is deeply conserved. Mouse ortholog *Cnot3* (NCBIGene:232791, chromosome 7); *Drosophila* ortholog *Not3*. NCBI Taxon: *Homo sapiens* (9606), *Mus musculus* (10090), *Drosophila melanogaster* (7227).
- **Natural disease in other species:** No naturally occurring companion-animal or wildlife analog of IDDSADF is catalogued (e.g., in OMIA). CNOT3 biology in animals is known almost entirely from engineered/conditional models (see §15), not spontaneous disease.
- **Comparative biology / conservation:** CNOT3's role as a conserved regulator of development and tissue identity is demonstrated across *Drosophila* and mouse (e.g., conserved regulator of heart function). This evolutionary conservation supports the human disease mechanism.

> *Kuba 2018:* "We had previously identified CNOT3, a scaffold subunit of the CCR4-NOT complex, as a conserved regulator of heart function in Drosophila and mouse." PMID: 29526923

• **Transmission / zoonosis:** Not applicable (genetic disorder).

15. Model Organisms

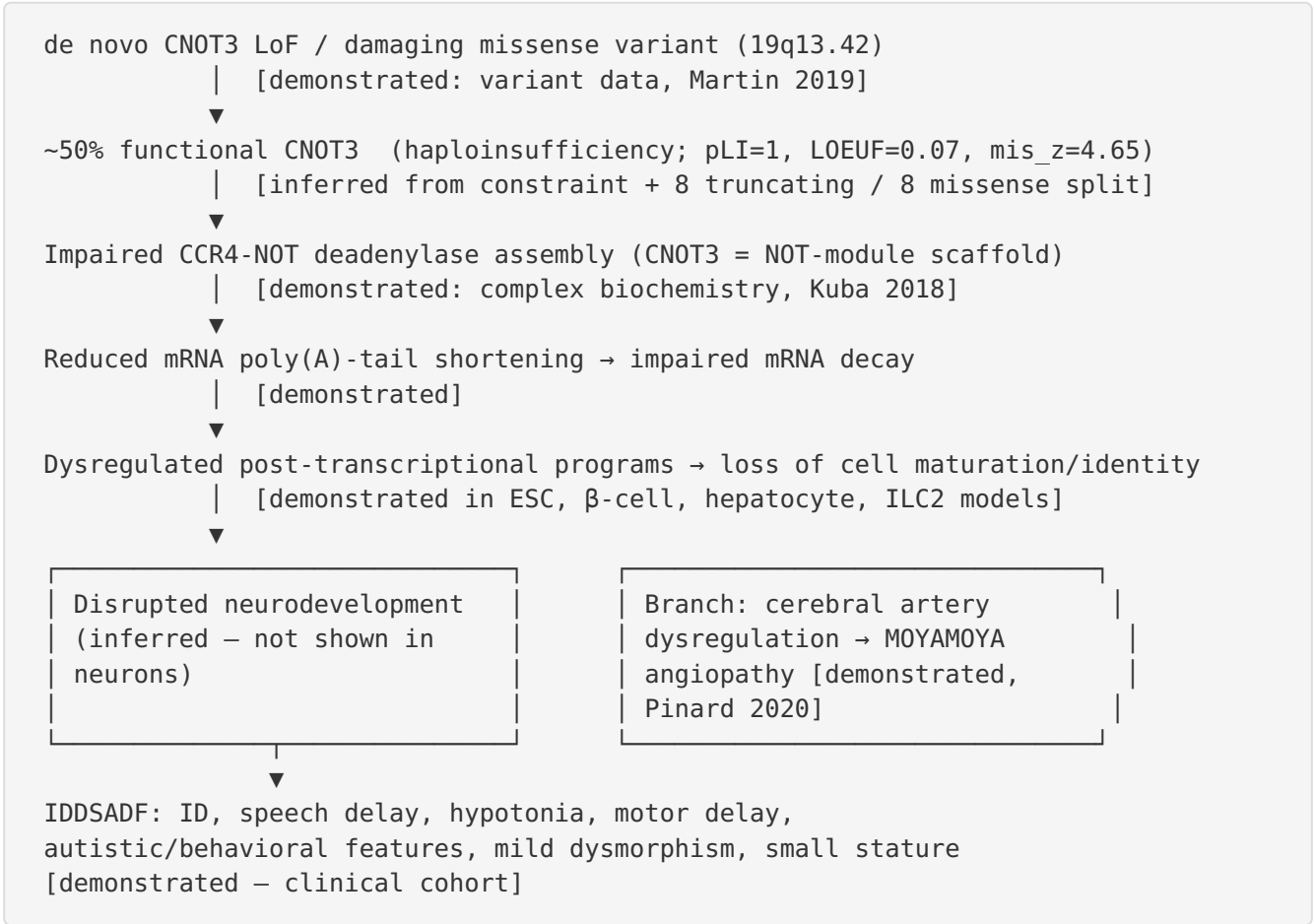
Cnot3 is **essential and evolutionarily conserved**; constitutive knockout is embryonic lethal, so disease modeling uses **conditional/tissue-specific** approaches. There is currently **no published neuron-specific or brain-conditional mouse model** directly recapitulating IDDSADF — a key gap.

Model system	Manipulation	Phenotype / finding	PMID
Mouse embryonic stem cells	*Cnot3* conditional deletion (CRISPR)	Disrupted pluripotency and differentiation; platform for dissecting CCR4-NOT in cell fate	37824068
Mouse β-cell–specific KO	*Cnot3* deletion in β-cells	Loss of β-cell identity/maturation, increased mRNA stability, diabetes	32859966
Mouse liver-specific KO (Cnot3-LKO)	*Cnot3* deletion in liver	Impaired hepatic maturation, hepatitis/necrosis, anemia, transient alopecia	31630801
Drosophila + mouse (heart)	*Cnot3/Not3* loss	Conserved regulator of heart function; links poly(A) shortening to *Atg7*/p53 transcriptional control	29526923
Mouse ILC2 conditional KO	*Cnot3* deletion in ILC2	Aberrant T-bet/RORγt; destabilization of *Tbx21*/*Rorc* — identity control	42118148

> *Suzuki et al. 2020:* "in mice lacking a Cnot3 gene, which encodes an essential component of the CCR4-NOT deadenylase complex in liver (Cnot3-LKO mice), the liver does not mature properly, resulting in various pathologies such as hepatitis, hepatic necrosis, and anemia." PMID: 31630801

Phenotype recapitulation. Across tissues the unifying theme is **failure of cell maturation/identity** upon *Cnot3* loss — strongly supporting the human haploinsufficiency mechanism — but **none of these models is neural**, so they recapitulate the *molecular* mechanism rather than the *neurodevelopmental phenotype*. **Model resources:** MGI (mouse *Cnot3*), FlyBase (*Not3*), IMPC/IMSR.

Mechanistic Model / Interpretation



The coherent story: a dosage-sensitive, ubiquitously required post-transcriptional regulator is reduced by ~half; this does not abolish any single pathway but subtly degrades the **fidelity and timing of cell-fate programs**. In the developing brain this manifests as a static encephalopathy; in cerebral vasculature it can later manifest as moyamoya. The strongest, directly demonstrated links are at the molecular/cellular level (steps 1–5); the neuron-specific translation of the defect (step 6) is currently **inferred by analogy** from non-neural models and from the clinical phenotype.

Evidence Base

PMID	Title (abbrev.)	Role	How it supports / challenges
31201375	*De novo variants in CNOT3 cause a variable neurodevelopmental disorder*	**Defining cohort**	Establishes gene–disease link (16 probands), phenotype spectrum, 8 truncating/8 missense, haploinsufficiency conclusion
32720325	*Inherited cases of CNOT3-associated IDDSADF*	Inheritance	First AD transmission; names IDDSADF/OMIM #618672; variable expressivity
31474762	*Pleiotropy… extends to moyamoya angiopathy*	Pleiotropy/ prognosis	CNOT3 *de novo* variants predispose to moyamoya — expands spectrum
29526923	*poly(A) shortening in heart function*	Mechanism	CNOT3 = CCR4-NOT scaffold; conserved *Drosophila*/mouse regulator
32859966	*Loss of β-cell identity… CNOT3 deadenylation*	Mechanism/ model	CNOT3-dependent deadenylation essential for maturation/identity
31630801	*Insufficient liver maturation (Cnot3-LKO)*	Model	Conditional mouse KO; Cnot3 essential, maturation failure
37824068	*Ccr4-Not in pluripotency and differentiation*	Model/ methods	ESC conditional deletion disrupts pluripotency/differentiation
39823890	*CNOT3 knockdown and senescence*	Mechanism	Links CNOT3 loss to p21/p53 mRNA stabilization and senescence
42118148	*CNOT3 supports ILC2 differentiation*	Mechanism	CNOT3 destabilizes *Tbx21*/*Rorc* — identity control

Constraint metrics (gnomAD) and ClinVar variant counts were queried directly during the investigation and corroborate the haploinsufficiency mechanism and the pathogenic-variant landscape.

Limitations and Knowledge Gaps

1. **Small evidence base.** The clinical phenotype rests largely on one 16-proband cohort plus scattered cases; frequencies are imprecise and prevalence is unknown.
2. **No neural model.** All functional *Cnot3* models are non-neural (ESC, β-cell, liver, heart, ILC2). The critical step linking CNOT3 haploinsufficiency to disrupted human neurodevelopment (step 6) is **inferred, not demonstrated**.
3. **No patient-tissue omics.** No transcriptomic/proteomic/methylation signature from patient neural tissue exists; there is no episignature and no biomarker.
4. **Genotype–phenotype correlation unresolved.** Variable expressivity (even intra-familial) is unexplained; modifier genes are unknown. It is unclear whether specific missense alleles are partial loss-of-function, dominant-negative, or truly equivalent to truncating alleles.
5. **Pleiotropy scope.** The penetrance and predictors of moyamoya in CNOT3 variant carriers are not quantified; surveillance guidelines are not evidence-based.
6. **No therapeutics.** No disease-specific trials, drugs, or outcome data.

Proposed Follow-up Experiments / Actions

1. **Generate a brain-conditional / heterozygous *Cnot3* mouse** (e.g., *Nestin-Cre* or neuron-specific heterozygous model) and phenotype for learning/memory, sociability, and cortical development — to directly test step 6 of the causal chain.
2. **Patient iPSC-derived neurons/organoids** with CNOT3 LoF and recurrent missense alleles; perform RNA-seq + poly(A)-tail profiling (nanopore direct RNA) to identify stabilized neural target mRNAs and build a candidate molecular signature.
3. **Allelic-series functional assays** (truncating vs recurrent missense) to classify missense alleles as LoF/hypomorph/dominant-negative and establish genotype–phenotype rules.
4. **Expand and deep-phenotype the cohort** via GeneMatcher/registry to refine HPO frequencies, define prevalence, and quantify moyamoya penetrance → inform surveillance guidelines.
5. **Search for a DNA-methylation episignature** in patient blood to enable VUS reclassification (as achieved for many chromatin-gene NDDs).
6. **Draft consensus management/surveillance recommendations**, including neurovascular imaging criteria for the moyamoya branch.

Consensus Answer

Intellectual Developmental Disorder with Speech Delay, Autism, and Dysmorphic Facies (IDDSADF; MONDO:0032864, OMIM #618672) is a rare autosomal-dominant Mendelian neurodevelopmental disorder caused by heterozygous, mostly *de novo* loss-of-function or damaging missense variants in **CNOT3** (19q13.42), a haploinsufficiency-sensitive scaffold subunit of the CCR4-NOT mRNA deadenylase complex. Reduced CNOT3 dosage impairs poly(A)-tail shortening and mRNA decay, dysregulating post-transcriptional programs that control cell differentiation and identity, producing variable intellectual disability/developmental delay with speech delay, hypotonia, motor delay, autistic/behavioral features, mild non-specific dysmorphism and small stature (seizures uncommon), with a subset developing moyamoya cerebrovascular disease. Management is entirely supportive with no targeted therapy, and diagnosis rests on trio exome/genome sequencing.